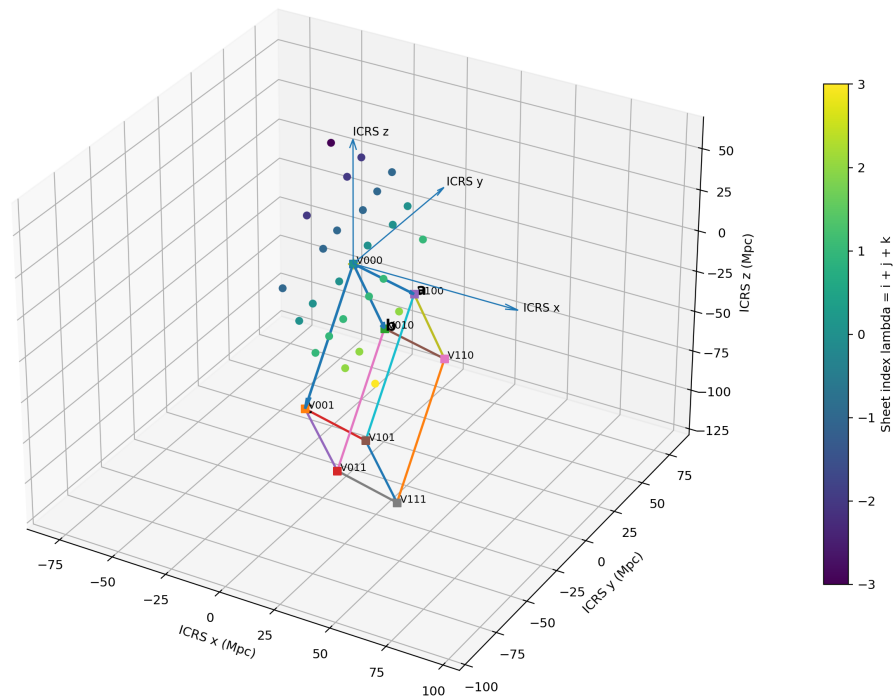


K-FIELD

Cell3 and Galaxy Shell Coherence

An ICRS-Oriented Unit-Sphere Calibration and Multiscale Shell Comparison

Calibrated Cell3 Base Cell: 8 Vertices and 27 Centered Half-Step Lucas Nodes



Title figure. The calibrated Cell3 base cell in ICRS coordinates, with all eight vertices and all 27 centered half-step Lucas nodes. The plotted coordinates are the independently calibrated Mpc realization of the unit-sphere natural Cell3 geometry.

Principal Investigator

P. Dwayne Esterline

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

Krytronx, LLC

Proprietary Confidential Information and Trade Secret of Krytronx, LLC.

All Rights Reserved. Do Not Distribute.

Publication epoch: 1784030762

Abstract

This work establishes a complete shell-coherence comparison between the observed radial distribution of 5,000 galaxies and a dense node population generated from the authoritative K-Field Cell3 geometry. The Cell3 geometry is retained in its correct International Celestial Reference System orientation, while the approximately 1.3436536 nanometre rendering magnitude is removed. The body diagonal is first normalized to the unit-sphere natural gauge, and the resulting shape is converted to a unit-volume calibration gauge. One scalar astronomical scale is then determined by the same shell-coherence procedure used to calibrate the galaxy-derived cell. The calibrated equivalent-volume scale is 46.1 Mpc, corresponding to a natural-gauge body diagonal of 126.318036607 Mpc and a reciprocal unit-volume scale of $0.021691973970 \text{ Mpc}^{-1}$. At depth 4, the construction produces 12,977,916 nodes in the radial domain, of which 6,884,597 survive the galaxy-survey sky mask. Spherical shell counts are evaluated at five shell half-widths, normalized for spherical area, and compared with the galaxy shell signal using Fourier spectra, cross-covariance, and four wavelet families. The full-band median wavelet coherence ranges from 0.5536 to 0.7978 across the tested families. Under an identical reconstruction and comparison method, the calibrated Cell3 geometry equals or exceeds the galaxy-derived cell in most complete-band and long-period coherence measures. The physical ellipsoid branch remains coherent but is more wavelet-family dependent. The combined result establishes a scale-selective, geometry-dependent relation between Cell3 shell organization and the observed galaxy shell population, with the spherical construction providing the clearest and most stable correspondence.

Foundational statement. The independently oriented Cell3 geometry, when expressed in the unit-sphere natural framework and calibrated by shell coherence rather than by its microscopic rendering scale, reproduces a strong multiscale relationship with the observed galaxy shell signal.

Scope and document contract

The paper is self-contained. It defines the ICRS frame, the Cell3 natural and unit-volume gauges, the survey mask, the depth-4 node construction, spherical and ellipsoidal shell statistics, scale calibration, covariance, Fourier analysis, and wavelet coherence. Every numerical table and every plotted result is generated from the stated Cell3 constants, the actual 5,000-galaxy dataset, or the archived output of the completed computational pipeline. The local 27-node figures describe the exact centered half-step node system of one Cell3 neighborhood; the multi-million-node shell analysis is a separate depth-4 continuation of the same basis.

Table of Contents

Abstract	2
Scope and document contract	2
1. The Scientific Question	6
1.1 Why Cell3 must be tested separately	6
1.2 Main result in one statement	6
2. Coordinate Frame, Empirical Data, and Survey Mask	7
2.1 ICRS Cartesian convention	7
2.2 Galaxy dataset	7
2.3 Equal-area-style angular mask	7
3. Authoritative Cell3 Geometry in the Unit-Sphere Natural Gauge	9
3.1 Removal of the microscopic rendering scale	9
3.2 Unit-volume calibration gauge	9
3.3 Calibrated Cell3 base cell	9
3.4 Primitive face planes and orientation	10
3.5 Seven-sheet node organization	11
4. From One Cell to the Depth-4 Astronomical Node Population	13
4.1 Dyadic continuation	13
4.2 Origin and phase	13
5. Spherical and Ellipsoidal Shell Signals	14
5.1 Spherical shell count	14
5.2 Area normalization and unit-peak display	14
5.3 Physical ellipsoid branch	15
6. Independent Astronomical Scale Calibration	16
6.1 Calibration principle	16
6.2 Coarse and fine sweeps	16
6.3 Meaning of the two scale gauges	17

7. Full-Length Wavelet Coherence	18
7.1 Why localized coherence is required	18
7.2 Coherence definition	18
7.3 Interpretation of the coherence field	19
8. Fourier Content and Rigid-Lag Covariance	20
8.1 Detrended Fourier spectra	20
8.2 Cross-covariance	21
9. Identical-Method Comparison with the Galaxy-Derived Cell	22
9.1 Why identical processing matters	22
9.2 Result of the geometry substitution	23
9.3 Relation to the earlier 3-4 percent dilation result	23
10. Ellipsoidal Shell Branch	24
11. Physical and Scientific Interpretation	25
11.1 What has been established	25
11.2 Why strong coherence can coexist with a noisy FFT	25
11.3 Why weak cross-covariance does not negate coherence	25
11.4 The calibrated Cell3 scale	25
12. New Discovery Pathways	26
12.1 Distance-uncertainty ensembles	26
12.2 Distance-method stratification	26
12.3 Larger and denser catalogs	26
12.4 Sector and directional analysis	26
12.5 Depth persistence	26
12.6 Cross-validation	26
13. Complete Reconstruction Protocol	27
14. Conclusions	28
Appendix A. Complete Cell3 Numerical Kernel	29

Appendix B. All 27 Centered Half-Step Lucas Nodes	30
Appendix C. Machine-Readable Evidence Files	32
Archival Notice	33

1. The Scientific Question

A geometric field theory makes a strong prediction when its internally generated spatial organization can be compared with an independently observed population. The K-Field shell experiment performs that comparison in radial space. The dense theoretical signal is formed by counting Cell3 lattice nodes in successive shells centered on the observer. The empirical signal is formed by counting catalog galaxies in the same radial shells. Their raw amplitudes cannot be compared directly because the node set contains millions of points while the galaxy sample contains only 5,000 objects. Their organization can be compared after matched masking, matched shell sampling, explicit normalization, and multiscale signal analysis.

The central question is not whether each galaxy occupies a uniquely identified Cell3 node. The shell test asks a population-level question: does the observed radial distribution carry scale content, localized oscillations, or repeated bands that correspond to the radial organization generated by the Cell3 geometry? This formulation is robust to sparse sampling and approximate individual distances because it compares collective structure rather than requiring exact object-to-node assignments.

1.1 Why Cell3 must be tested separately

The first galaxy shell experiments used a direct cell reconstructed from galaxy sky geometry. That cell supplied a macroscopic, data-derived coordinate system. Cell3 is different. It is the authoritative K-Field primitive geometry, with its own shape, handedness, plane system, sheet normal, and fixed ICRS orientation. A rigorous comparison therefore requires replacing the galaxy-derived basis with Cell3 while holding the rest of the pipeline fixed. The absolute nanometre rendering length cannot be carried into the astronomical experiment. The Cell3 shape and orientation must instead be expressed in a dimensionless natural gauge and assigned one independently fitted astronomical scale.

1.2 Main result in one statement

The calibrated Cell3 equivalent-volume scale is 46.1 Mpc. With this scale, the survey-masked Cell3 shell signal exhibits broad wavelet coherence with the 5,000-galaxy shell signal. The strongest full-band median is 0.7978 for the complex Gaussian order-2 wavelet. The Cell3 geometry produces higher complete-band median coherence than the galaxy-derived cell in all four tested wavelet families, and it remains stronger in three of four comparisons when the analysis is restricted to periods of at least 2 Mpc.

2. Coordinate Frame, Empirical Data, and Survey Mask

2.1 ICRS Cartesian convention

All direct vectors are expressed as ICRS Cartesian columns. The x axis points toward right ascension 0 degrees and declination 0 degrees. The y axis points toward right ascension 90 degrees and declination 0 degrees. The z axis points toward the north celestial pole. For any nonzero vector $v=(x,y,z)$, right ascension and declination are recovered by $RA=\text{atan2}(y,x)$ and $Dec=\text{asin}(z/|v|)$.

$$u(\alpha, \delta) = [\cos(\delta) \cos(\alpha), \cos(\delta) \sin(\alpha), \sin(\delta)]^T$$

2.2 Galaxy dataset

The empirical witness contains 5,000 galaxies with catalog right ascension, declination, and distance. The radial domain is 13.327534876 to 218.674365624 Mpc. These distances are heterogeneous astronomical estimates and are not treated as exact rulers. The shell test instead uses their aggregate population signal. At a shell half-width of 0.125 Mpc, the narrow shell signal averages only a few galaxy observations per radial sample, making the galaxy series sparse relative to the theoretical node series.

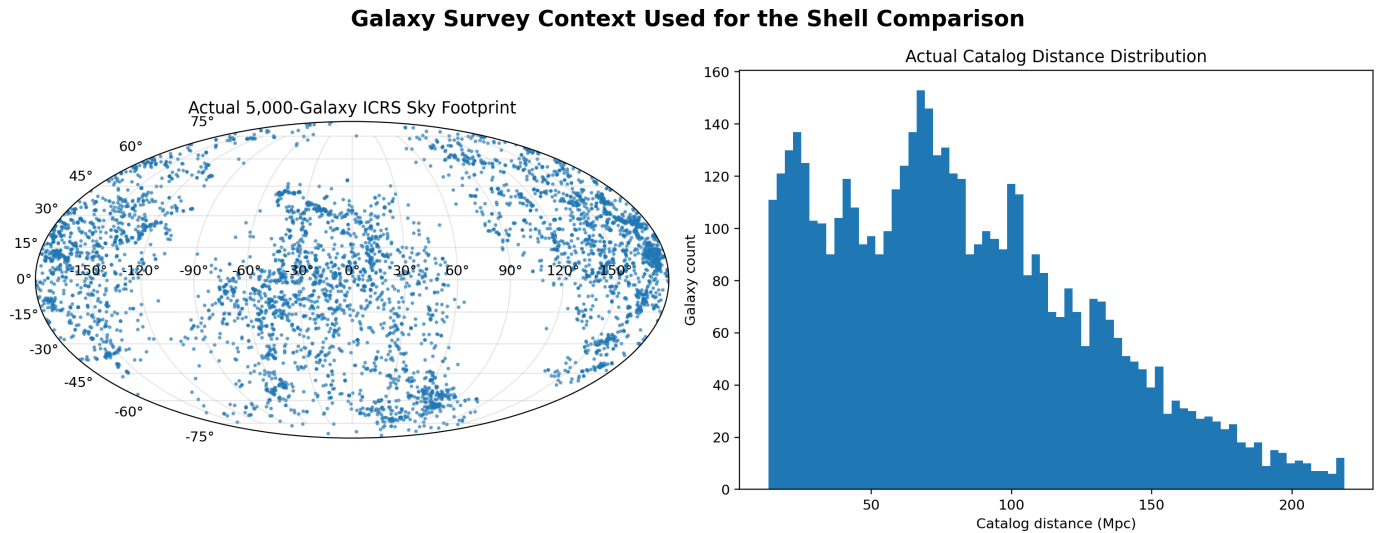


Figure 1. Actual sky footprint and catalog-distance distribution of the 5,000 galaxies used in the comparison. No synthetic objects are added.

2.3 Equal-area-style angular mask

The K-Field node population is not compared over the entire sphere when the galaxy survey does not occupy the entire sphere. The observed galaxy directions are binned into 72 right-ascension intervals and 36 equal intervals of $\mu=\sin(\text{Dec})$. This produces angular bins of equal solid angle before occupancy is applied. A bin is retained if at least one catalog galaxy occupies it. The same occupied-bin mask is applied to every theoretical node direction. This operation makes the angular support of the two radial signals comparable while preserving the full galaxy sample.

Quantity	Value	Interpretation
RA bins	72	Five-degree azimuthal sampling
$\sin(\text{Dec})$ bins	36	Equal-solid-angle latitudinal sampling
Galaxy records	5,000	Observed shell population
Cell3 node depth	4	Dyadic denominator 16
All Cell3 nodes in radial domain	12,977,916	Before sky masking
Survey-masked Cell3 nodes	6,884,597	Used in matched shell sweep

Table 1. Dataset and survey-mask contract.

3. Authoritative Cell3 Geometry in the Unit-Sphere Natural Gauge

3.1 Removal of the microscopic rendering scale

The authoritative Cell3 basis is supplied as three ICRS-oriented edge vectors. Their numerical rendering has a body-diagonal magnitude of 1.343653621680 in the microscopic representation. That value is not used as an astronomical scale. The complete basis is divided by the norm of $a+b+c$, forcing the natural body diagonal to unity while preserving every direction, edge ratio, angle, face relation, determinant sign, and ICRS orientation.

$$B_{\text{natural}} = B_{\text{render}} / ||a_{\text{render}} + b_{\text{render}} + c_{\text{render}}||$$

$$||a_{\text{natural}} + b_{\text{natural}} + c_{\text{natural}}|| = 1$$

Basis	x	y	z	Length
a natural	0.243739336378	-0.040856517632	-0.027899677634	0.248709692499
b natural	0.026294556331	0.154295606469	-0.407397469526	0.436430104420
c natural	-0.007538637535	-0.290095375828	-0.513327540100	0.589675607057

Table 2. Cell3 edge vectors after unit-sphere natural normalization.

3.2 Unit-volume calibration gauge

The natural Cell3 determinant is 0.0486078795553. To separate shape from one free dimensional scale, the basis is divided by the cube root of this determinant magnitude. The resulting unit-volume gauge has determinant magnitude one. A scalar S in Mpc can then be applied to every basis vector. Because the unit-volume cell has volume one, the calibrated physical cell volume is exactly S cubed.

$$v_{\text{root}} = |\det(B_{\text{natural}})|^{(1/3)} = 0.364951840910$$

$$A_{\text{unit-volume}} = B_{\text{natural}} / v_{\text{root}}$$

$$|\det(A_{\text{unit-volume}})| = 1$$

$$B_{\text{physical}}(S) = S A_{\text{unit-volume}}$$

Basis	x	y	z
a unit-volume	0.667867123975	-0.111950435790	-0.076447559667
b unit-volume	0.072049386749	0.422783472153	-1.116304739031
c unit-volume	-0.020656526945	-0.794886731095	-1.406562407850

Table 3. Unit-volume Cell3 basis used for astronomical scale calibration.

3.3 Calibrated Cell3 base cell

Quantity	Value
Equivalent-volume scale S	46.100000 Mpc
Natural body-diagonal scale	126.318036607 Mpc
Cell volume	97972.181000 Mpc ³
$ a $	31.416520042 Mpc
$ b $	55.128993907 Mpc
$ c $	74.486664919 Mpc
angle(a,b)	83.933483012 deg
angle(a,c)	80.448130018 deg
angle(b,c)	50.363231814 deg
Reciprocal unit-volume scale	0.021691973970 Mpc ⁻¹
Reciprocal natural scale	0.007916525833 Mpc ⁻¹

Table 4. Calibrated physical dimensions derived from the unit-volume Cell3 gauge.

Calibrated Cell3 Base Cell: 8 Vertices and 27 Centered Half-Step Lucas Nodes

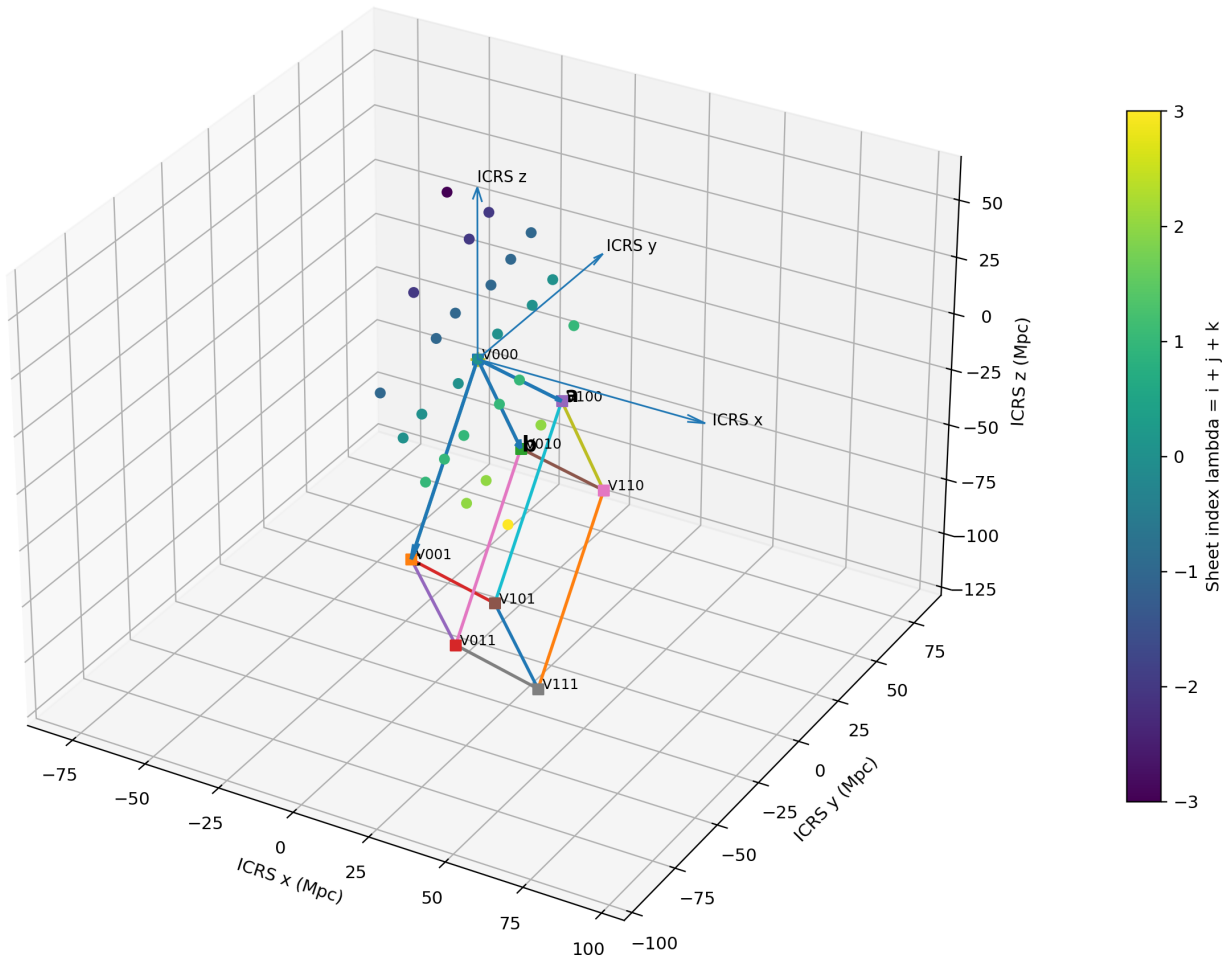


Figure 2. Dimensionally correct calibrated Cell3 base cell. The eight corner vertices are labeled V000 through V111. The 27 centered half-step Lucas nodes are generated from $r=B(i,j,k)/2$ for i,j,k in $\{-1,0,1\}$. Color records the seven constant-sum sheets $\lambda=i+j+k$.

3.4 Primitive face planes and orientation

Plane	Area (Mpc ²)	Normal x	Normal y	Normal z	RA	Dec
P_ab	1722.261941	0.194092	0.913177	0.358379	78.001 deg	21.001 deg
P_ac	2307.667975	-0.089053	-0.866577	0.491034	264.133 deg	29.409 deg
P_bc	3162.335867	0.995965	-0.083602	0.032619	355.202 deg	1.869 deg

Table 5. Primitive Cell3 face planes in the calibrated ICRS geometry. Normal signs are chosen for display; each plane is projective.

Primitive Cell3 Face Planes in the Calibrated ICRS Geometry
 P_ab: area=1722.262 Mpc² P_ac: area=2307.668 Mpc² P_bc: area=3162.336 Mpc²

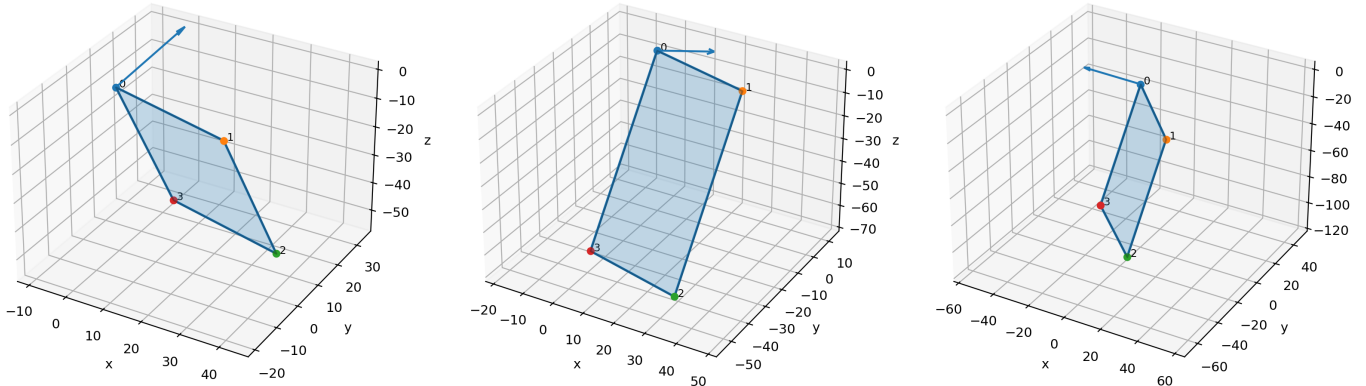


Figure 3. Primitive calibrated face planes P_ab, P_ac, and P_bc, with exact parallelogram corners and normals derived from cross products of the Cell3 edge vectors.

3.5 Seven-sheet node organization

The centered half-step node system contains 27 positions and seven sheet values. For the integer address $L=(i,j,k)$, the local node is $r=B L/2$ and the sheet coordinate is $\lambda=i+j+k$. The sheet normal is proportional to $B^{(-T)}(1,1,1)$. In the calibrated ICRS geometry, its displayed projective direction is RA 183.082265 degrees and Dec 28.568338 degrees.

Seven Constant-Sum Cell3 Sheets Through the 27 Centered Nodes

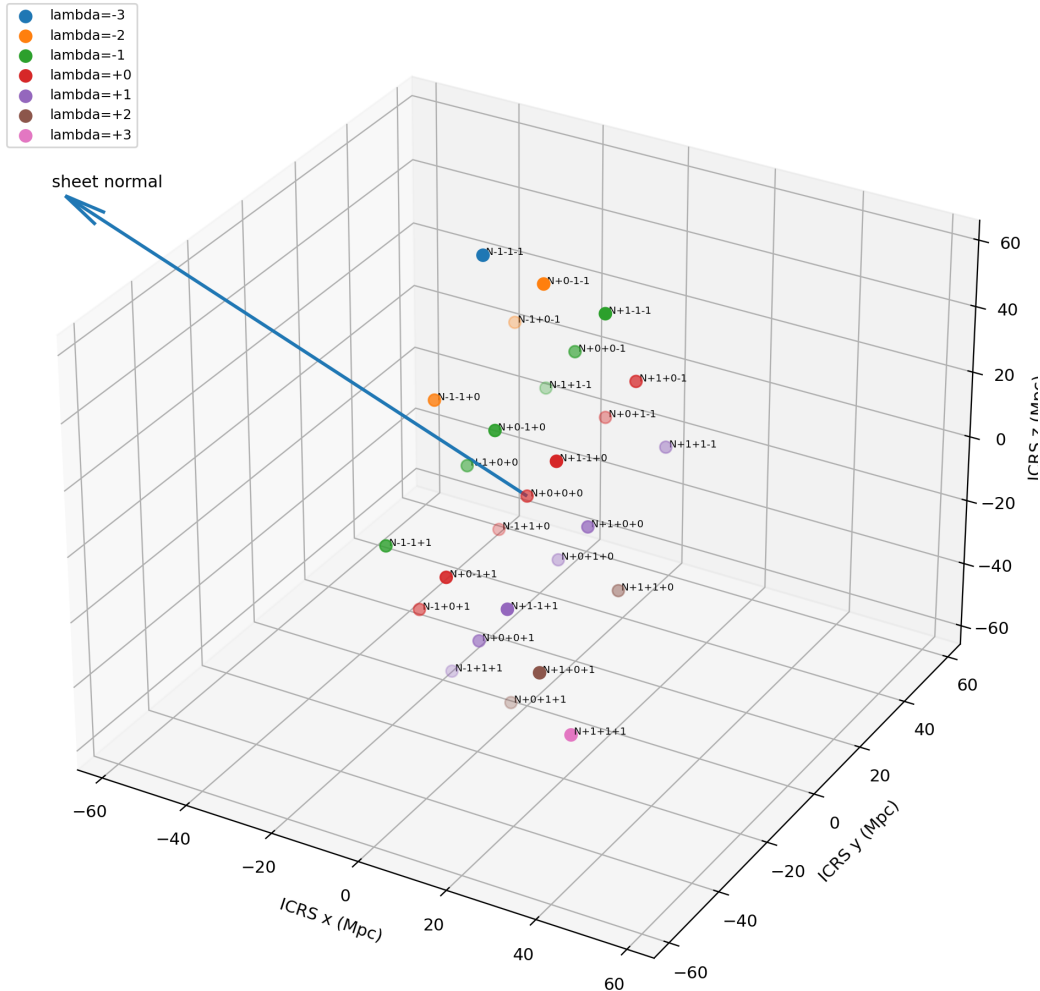


Figure 4. All 27 centered half-step nodes separated into the seven constant-sum sheets. Each node label records its signed integer triplet. The sheet-normal arrow is derived from the calibrated direct basis.

4. From One Cell to the Depth-4 Astronomical Node Population

4.1 Dyadic continuation

The local 27-node figure is the minimal centered half-step neighborhood. The astronomical shell signal uses a much denser depth-4 continuation. Each hierarchy coordinate is a multiple of 1/16. With zero Cell3 phase, a depth-4 address $m=(m_a, m_b, m_c)$ maps to physical ICRS position $x=S \text{ A_unit-volume } (m/16)$. Integer ranges are chosen large enough to cover the full galaxy radial domain. Nodes outside the domain are discarded, and the survey mask is then applied to the remaining directions.

```
q = m / 16
x_mpc = S A_unit-volume q
r = ||x_mpc||
keep if 13.327534876 <= r <= 218.674365624 Mpc
```

The completed calculation contains 12,977,916 depth-4 radii before angular masking and 6,884,597 radii after the survey mask. The theoretical signal is consequently dense enough to resolve narrow shell modulation, while the galaxy signal remains shot-noise limited.

4.2 Origin and phase

The Cell3 lattice is centered on the ICRS origin with phase $p=(0,0,0)$. The phase vector previously fitted for the galaxy-derived coordinate system is not imported. This distinction is essential: the experiment substitutes Cell3 geometry as an independent object rather than deforming Cell3 into the previously fitted galaxy hierarchy.

5. Spherical and Ellipsoidal Shell Signals

5.1 Spherical shell count

For shell center r and half-width T , the spherical count is the number of retained objects whose Euclidean radius lies within $[r-T, r+T]$. The same definition is applied to galaxies and masked Cell3 nodes. Shell centers advance in 0.05 Mpc increments across the complete radial domain. Five half-widths are used: 0.125, 0.25, 0.5, 1.0, and 2.0 Mpc.

$$N_T(r) = \sum_n 1(||x_n|| - r | \leq T)$$

5.2 Area normalization and unit-peak display

A spherical shell at larger radius covers more surface area. The count is therefore divided by $4\pi r^2$ before comparing radial shape. For signal analysis, each series is then centered, standardized, or normalized to its own mean as required by the method. For visual overlays, each displayed curve can also be divided by its own maximum so that both occupy the same y-axis range without implying equal population density.

$$\begin{aligned} \rho_T(r) &= N_T(r) / (4 \pi r^2) \\ y_T(r) &= \rho_T(r) / \text{mean}_r[\rho_T(r)] \\ y_{\text{peak}}(r) &= y_T(r) / \text{max}_r[y_T(r)] \end{aligned}$$

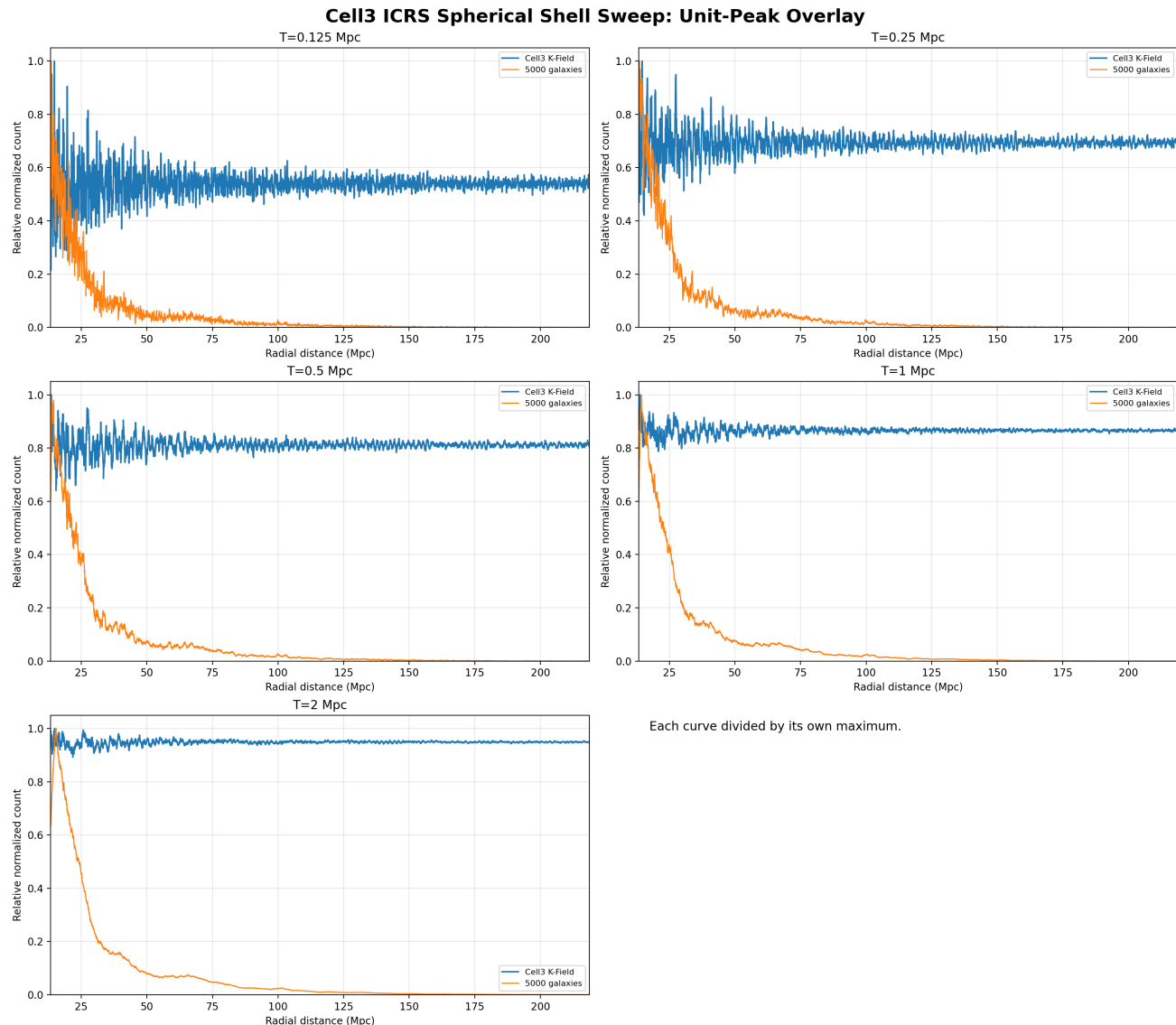


Figure 5. Unit-peak normalized spherical shell overlays for the five shell half-widths. The galaxy and Cell3 series are normalized independently so the figure compares radial form rather than raw population size.

5.3 Physical ellipsoid branch

The Cell3 direct basis also defines a quadratic family of homothetic ellipsoids. In the unit-volume gauge, $H=(A A^T)^{-1}$. The ellipsoidal radius is $\rho_E(x)=\sqrt{x^T H x}$. The ellipsoidal experiment applies the same scale, radial sampling, survey mask, and wavelet calculations, while replacing ordinary spherical radius with this anisotropic coordinate.

$$H = (A_{\text{unit-volume}} A_{\text{unit-volume}}^T)^{-1}$$

$$\rho_E(x) = \sqrt{x^T H x}$$

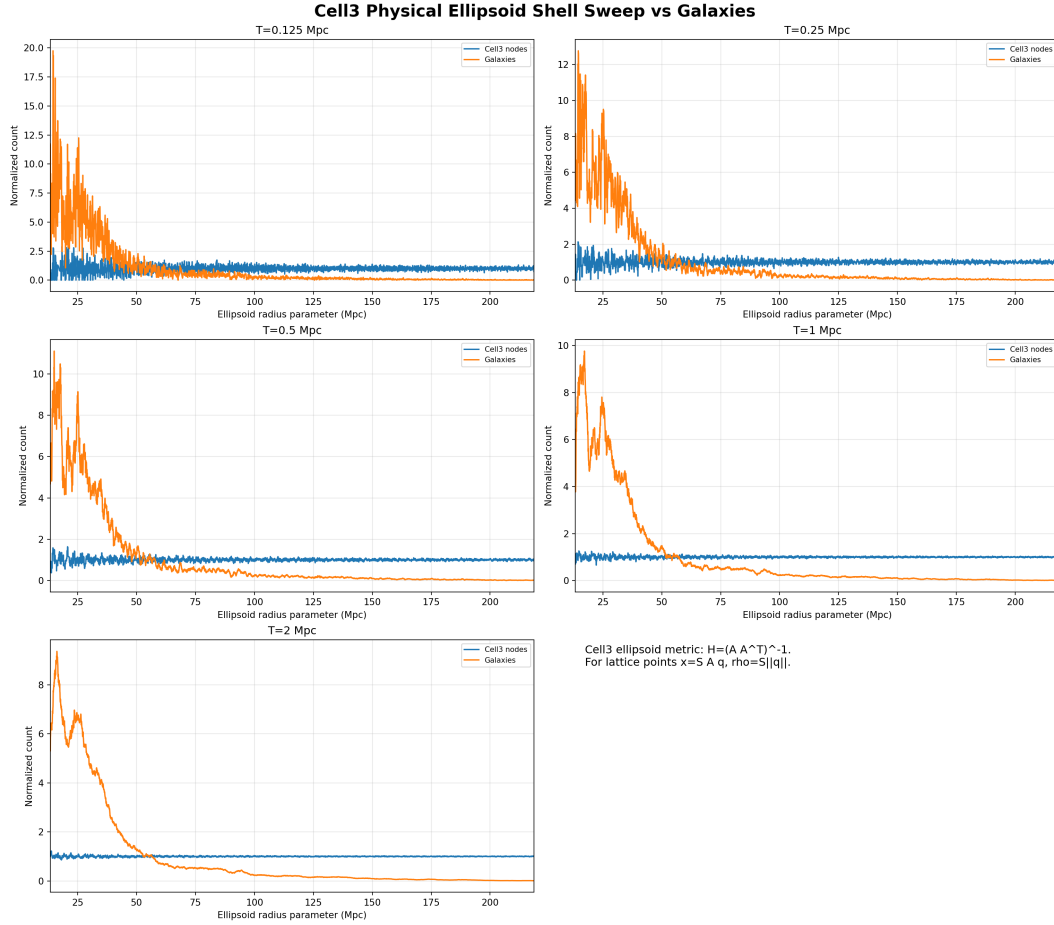


Figure 6. Galaxy and calibrated Cell3 signals evaluated on the physical Cell3 ellipsoid family. The same catalog, mask, radial grid, and normalization are retained.

6. Independent Astronomical Scale Calibration

6.1 Calibration principle

Cell3 enters the experiment with fixed dimensionless shape and fixed ICRS orientation but no astronomical length. Calibration therefore changes one scalar S and regenerates the spherical node signal at each candidate value. The galaxy signal is held fixed. For every candidate, wavelet coherence is calculated over the full radial overlap using four wavelet families. The preferred scale is selected by a consensus statistic rather than by one isolated Fourier peak or one wavelet.

6.2 Coarse and fine sweeps

The coarse search spans 26 to 50 Mpc in equivalent-volume scale. It identifies the high-coherence neighborhood near 46 Mpc. A refined four-wavelet sweep resolves the consensus maximum at 46.1 Mpc. The fine search is repeated around the leading coarse neighborhoods to ensure that the reported optimum is not a single-grid artifact.

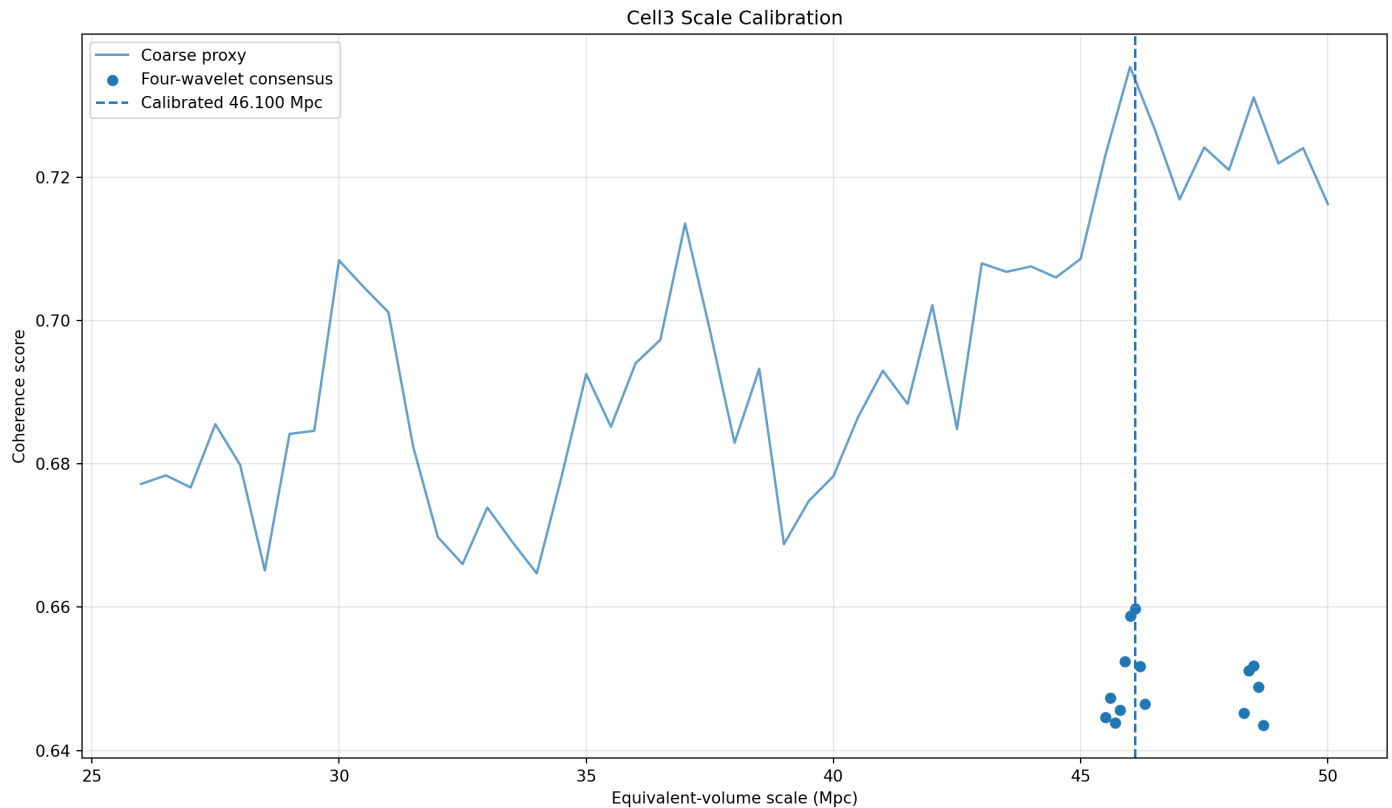


Figure 7. Cell3 scale calibration. The continuous coarse proxy identifies candidate neighborhoods; the discrete points are the completed four-wavelet consensus evaluations. The dashed line marks the calibrated equivalent-volume scale of 46.1 Mpc.

Calibration quantity	Value
Coarse search interval	26-50 Mpc
Coarse peak	46.0 Mpc
Fine consensus peak	46.1 Mpc
Natural-gauge body diagonal	126.318036607 Mpc
κ , unit-volume gauge	$0.021691973970 \text{ Mpc}^{-1}$
κ , natural gauge	$0.007916525833 \text{ Mpc}^{-1}$
Galaxy-derived equivalent-volume scale	36.932019638 Mpc
Cell3 / galaxy-cell scale ratio	1.248239345
Equivalent-volume difference	24.823934%

Table 6. Scale-calibration outcome. The 24.82 percent comparison refers to equivalent-volume gauge scales, not to microscopic Cell3 length.

6.3 Meaning of the two scale gauges

The value 46.1 Mpc is the scalar multiplying a determinant-one Cell3 basis. The value 126.318036607 Mpc is the scalar multiplying the original unit-sphere natural basis whose body diagonal equals one. They describe the same physical lattice. The conversion follows from the determinant cube root 0.364951840910. Neither value uses the nanometre rendering scale.

7. Full-Length Wavelet Coherence

7.1 Why localized coherence is required

A sparse galaxy signal does not present a stable stationary spectrum over the full radial interval. Individual catalog distances broaden narrow shell peaks, and the number of galaxies per shell changes strongly with radius. A Fourier transform averages all radii into one global spectrum. Wavelet coherence retains both radial location and period, allowing common bands to appear even when they are intermittent or phase-localized.

7.2 Coherence definition

For wavelet transforms W_g and W_K of the galaxy and Cell3 shell signals, magnitude-squared wavelet coherence compares the smoothed cross-wavelet power with the product of the smoothed individual powers. Values range from zero to one. Absolute values depend on the wavelet family and smoothing rule; the stable conclusions are the location of the calibrated scale, persistence across wavelets, and identical-method comparison between geometries.

$$R^2(r, p) = \frac{|S(W_g(r, p) W_K(r, p)^*)|^2}{[S(|W_g(r, p)|^2) S(|W_K(r, p)|^2)]}$$

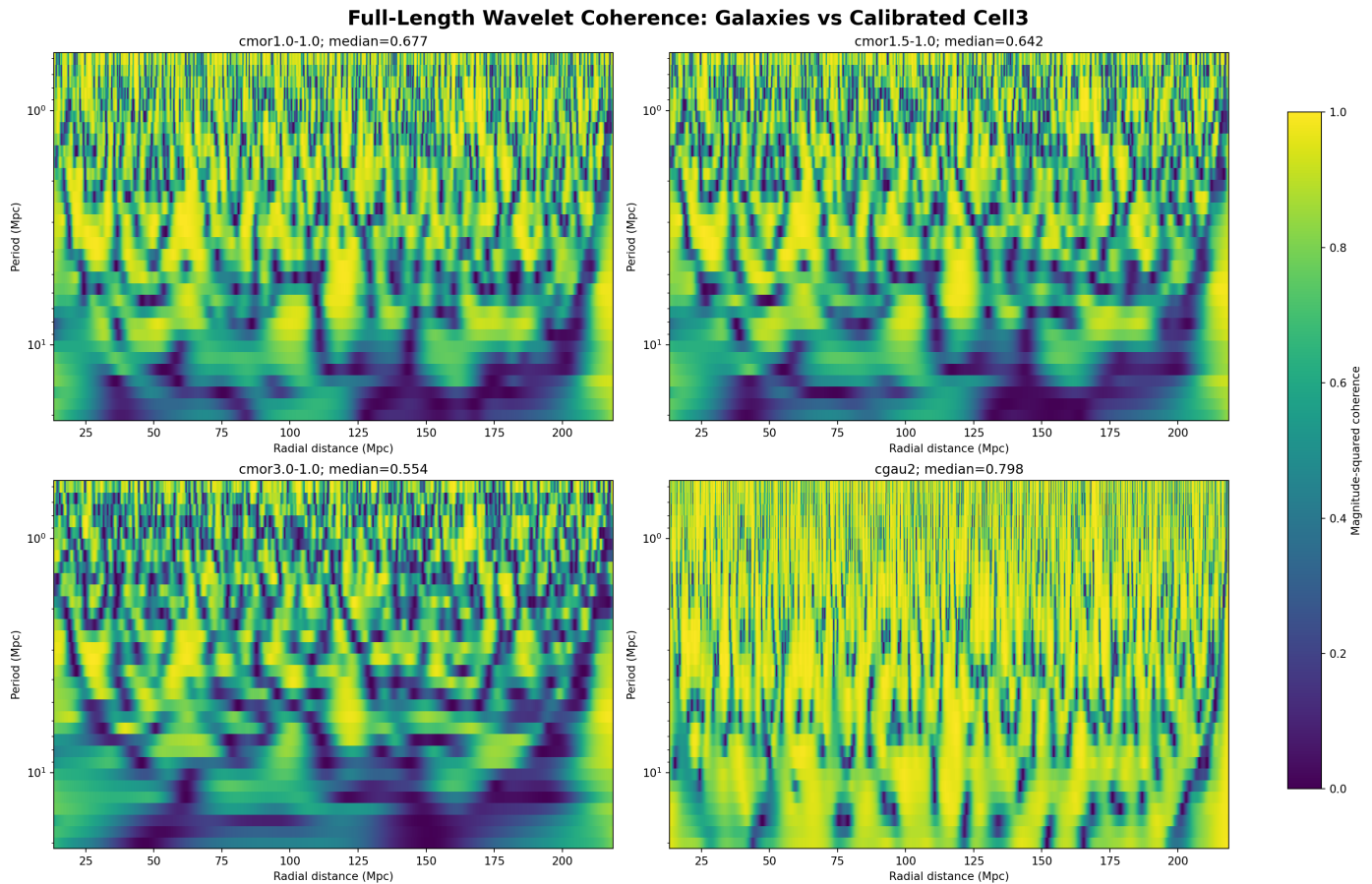


Figure 8. Full-length magnitude-squared wavelet coherence between the 5,000-galaxy spherical shell signal and the calibrated, survey-masked Cell3 signal. The four panels use three complex Morlet bandwidths and one complex Gaussian wavelet.

Wavelet	Full-band median	Leading-peak mean	Maximum	Period at maximum
cmor1.0-1.0	0.677249	0.732389	0.777018	0.600 Mpc
cmor1.5-1.0	0.642292	0.711038	0.777209	0.600 Mpc
cmor3.0-1.0	0.553626	0.643006	0.739245	0.600 Mpc
cgau2	0.797834	0.826220	0.836666	0.600 Mpc

Table 7. Spherical Cell3-galaxy wavelet-coherence summary at the calibrated scale and $T=0.125$ Mpc.

7.3 Interpretation of the coherence field

The coherence maps contain broad high-coherence regions rather than one isolated line. The complex Gaussian result has a full-band median of 0.7978 and remains strong through the long-period range. The Morlet medians decrease as the time-bandwidth localization becomes more restrictive, but they remain between 0.5536 and 0.6772. The agreement across distinct wavelet families demonstrates that the result is not produced by one selected mother wavelet. The strongest absolute values occur near the lower tested period boundary for most families, so complete-band medians and long-period medians are more conservative than the single maximum.

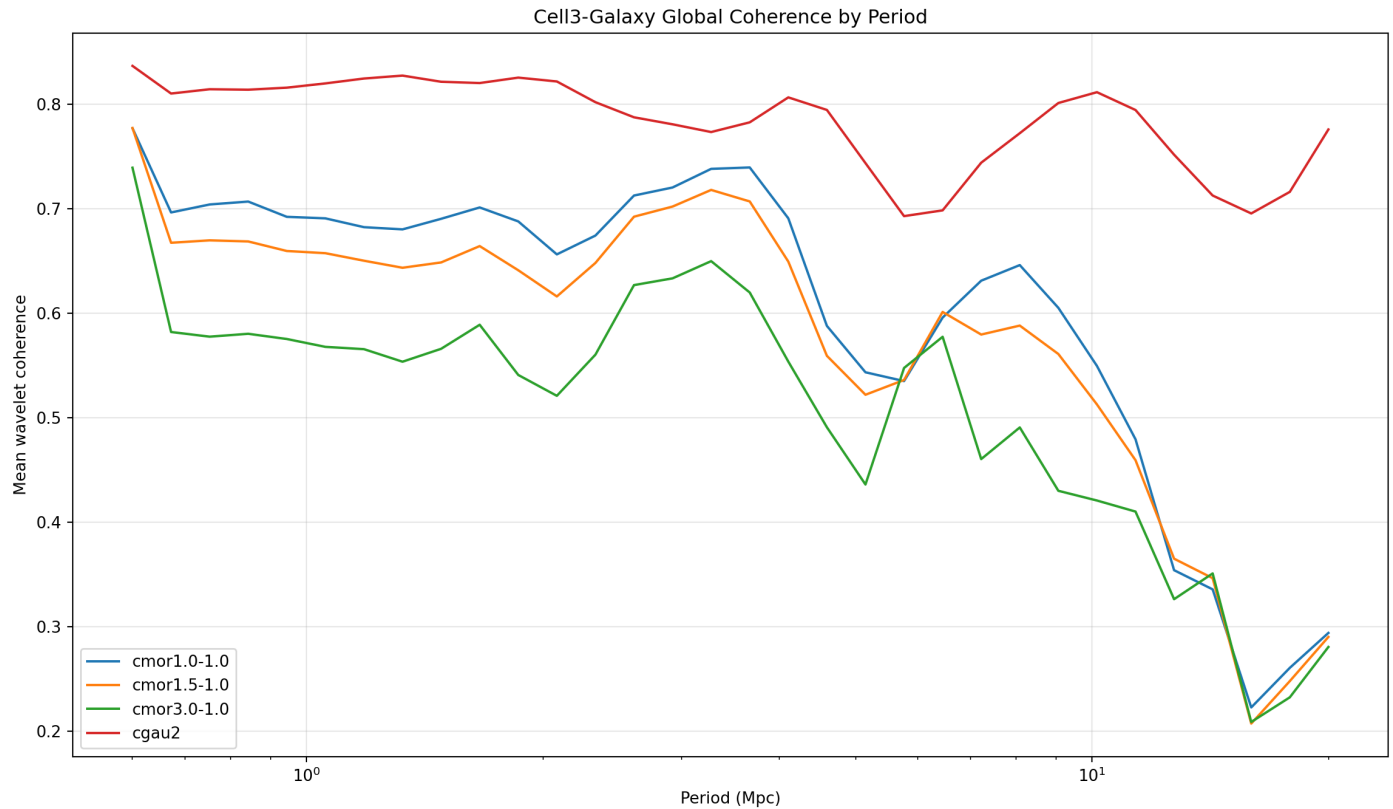


Figure 9. Radially averaged coherence as a function of physical period. This view separates period-dependent structure from its radial localization in Figure 8.

8. Fourier Content and Rigid-Lag Covariance

8.1 Detrended Fourier spectra

Fourier analysis is applied after removing a broad radial trend and multiplying each residual series by a Hann window. The transform identifies stationary period content that persists over the complete radial interval. Because the galaxy signal is sparse, the Fourier spectrum is not expected to reproduce every Cell3 peak. The informative result is the presence of exact or nearby period families that are also visible in the localized wavelet analysis.

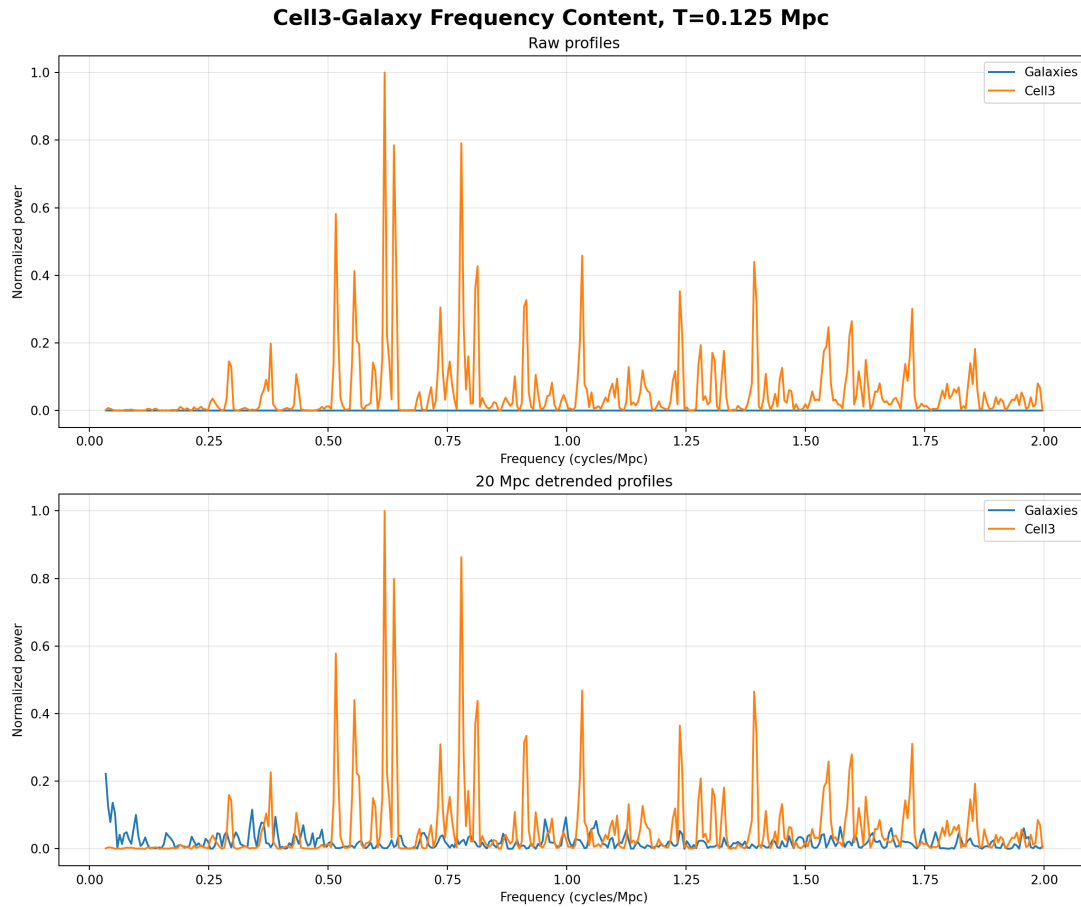


Figure 10. Detrended Fourier spectra of the galaxy and calibrated Cell3 spherical shell signals. The frequency and period views are derived from the same 0.05 Mpc radial sampling.

signal	rank	frequency_cycles_per_mpc	period_mpc	normalized_power
galaxy	1	0.053554	18.6727	0.000164447
galaxy	2	0.340798	2.93429	0.000124026
galaxy	3	0.998053	1.00195	9.36383e-05
galaxy	4	0.389484	2.5675	9.17361e-05
galaxy	5	0.097371	10.27	9.08653e-05
galaxy	6	0.954236	1.04796	8.66806e-05
galaxy	7	1.06134	0.942202	8.18719e-05
galaxy	8	0.360273	2.77568	7.33916e-05
galaxy	9	0.447907	2.23261	7.01034e-05
galaxy	10	1.57254	0.635913	6.38752e-05
galaxy	11	1.95716	0.510945	6.10107e-05
galaxy	12	1.67965	0.595362	6.05895e-05

Table 8. Leading archived Fourier peaks and nearest-period comparisons. Column names retain the machine-readable analysis terminology.

Principal correspondences include an exact FFT-bin match at 0.808661 Mpc, a 2.75 percent separation between 0.968868 and 0.942202 Mpc, a 4.08 percent separation between 1.092553 and 1.047959 Mpc, and a broader family near 1.94-2.05 Mpc. These matches do not imply that the full spectra are identical. They identify repeated scales within a much noisier empirical signal.

8.2 Cross-covariance

Cross-covariance asks whether one complete radial series is primarily a translated version of the other. The answer is no. At the narrowest shell width, the strongest normalized positive value is 0.04288 at -0.25 Mpc, while the strongest negative value is -0.09646 at -1.55 Mpc. The weak rigid-lag relation is compatible with strong wavelet coherence because localized bands can agree without requiring one globally fixed displacement.

T (Mpc)	Zero lag	Maximum	Lag at max	Minimum	Lag at min
0.125	-0.020360	0.042882	-0.25 Mpc	-0.096457	-1.55 Mpc
0.250	-0.005280	0.018385	-0.20 Mpc	-0.103154	-1.55 Mpc
0.500	-0.012226	0.017097	6.35 Mpc	-0.130079	-6.75 Mpc
1.000	-0.053240	-0.001889	11.55 Mpc	-0.240280	-5.70 Mpc
2.000	-0.104096	-0.030177	14.10 Mpc	-0.317387	-5.30 Mpc

Table 9. Normalized spherical cross-covariance summary.

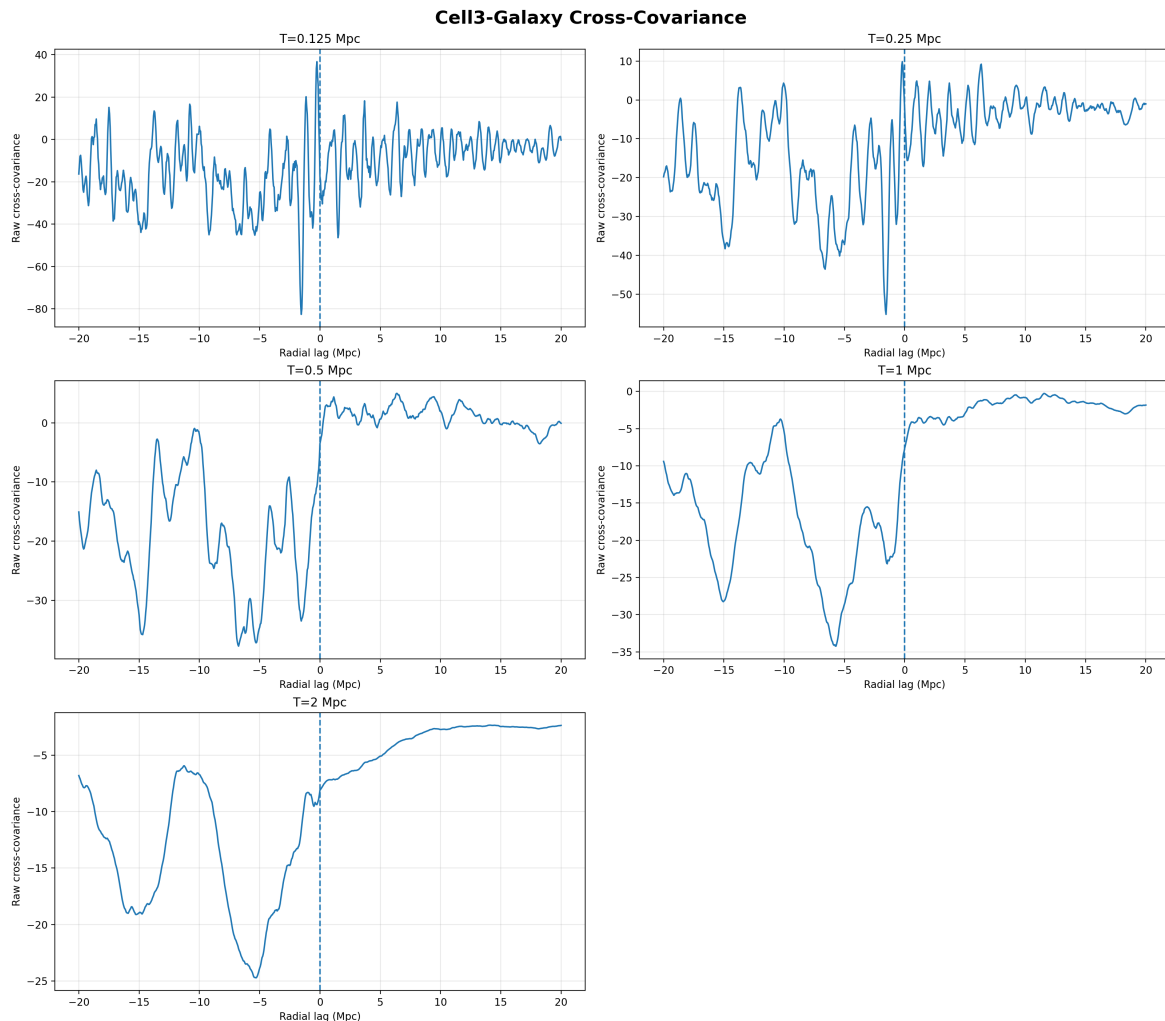


Figure 11. Cross-covariance as a function of physical radial lag for all five shell half-widths.

9. Identical-Method Comparison with the Galaxy-Derived Cell

9.1 Why identical processing matters

A geometry comparison is meaningful only when the computational treatment is the same. Cell3 and the galaxy-derived cell are therefore evaluated with the same 5,000 galaxies, radial domain, 0.05 Mpc step, 0.125 Mpc shell half-width, sky mask, depth-4 node construction, normalization, wavelet periods, smoothing, and summary statistics. Only the direct cell geometry, phase convention, and independently calibrated scalar differ.

Geometry	Wavelet	Median 0.6-20	Median 1-20	Median 2-20	Maximum	Period at max
Cell3	cgau2	0.797834	0.787472	0.775826	0.836666	0.600
Cell3	cmor1.0-1.0	0.677249	0.646038	0.595802	0.777018	0.600
Cell3	cmor1.5-1.0	0.642292	0.601204	0.560941	0.777209	0.600
Cell3	cmor3.0-1.0	0.553626	0.540882	0.490661	0.739245	0.600
Galaxy-derived cell	cgau2	0.776411	0.768456	0.760248	0.862666	2.923
Galaxy-derived cell	cmor1.0-1.0	0.629246	0.610656	0.593611	0.779618	0.600
Galaxy-derived cell	cmor1.5-1.0	0.612795	0.592194	0.562705	0.769464	0.600
Galaxy-derived cell	cmor3.0-1.0	0.537258	0.527234	0.489779	0.719746	0.600

Table 10. Cell3 and galaxy-derived cell compared under an identical wavelet-coherence method.

Identical Wavelet-Coherence Comparison: Galaxy-Derived Cell vs Cell3

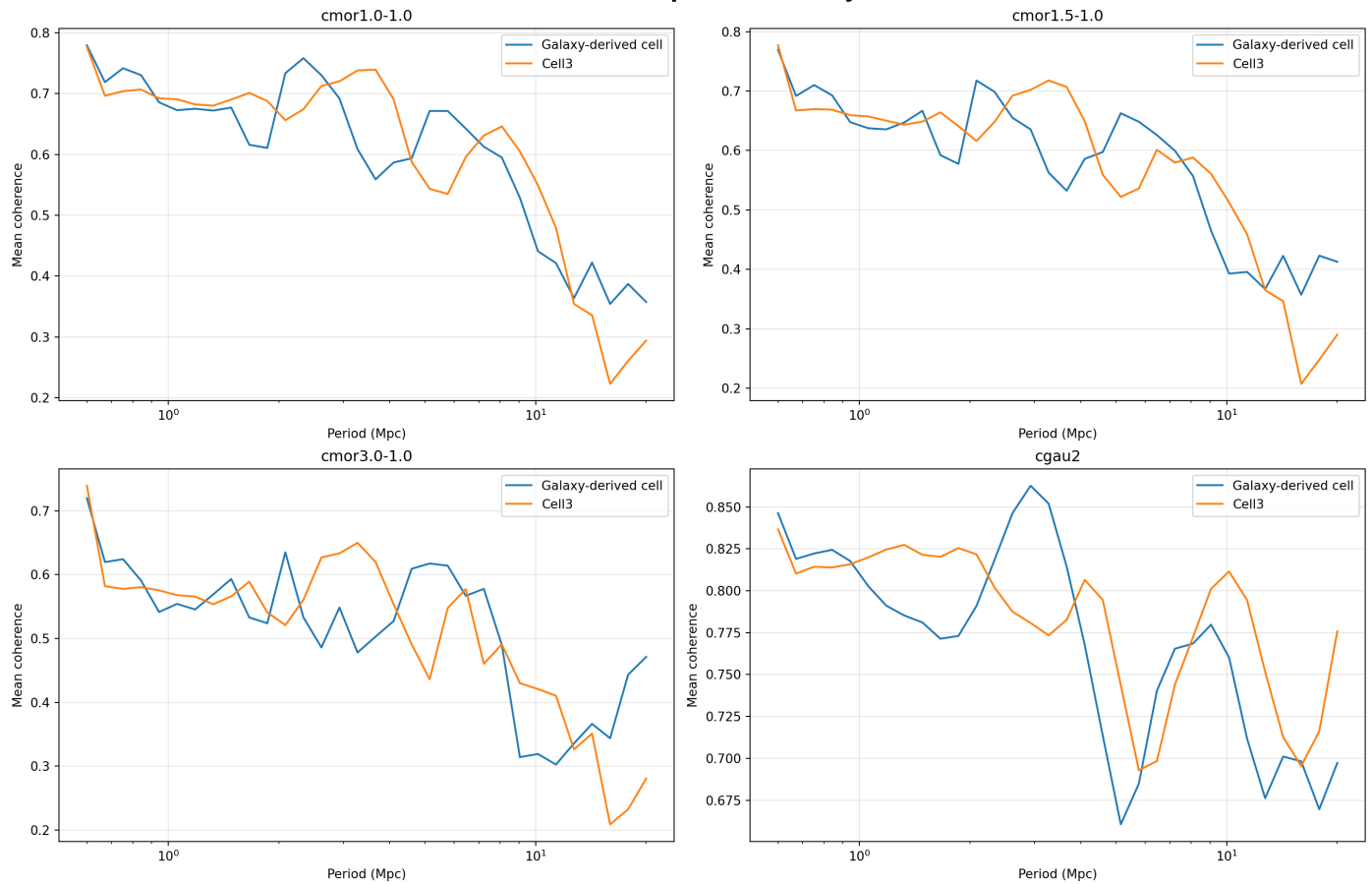


Figure 12. Direct wavelet-coherence comparison between calibrated Cell3 and the previously calibrated galaxy-derived cell. Cell3 is higher in all four complete-band medians and in three of four medians for periods at least 2 Mpc.

9.2 Result of the geometry substitution

The complex Gaussian median increases from 0.7764 for the galaxy-derived cell to 0.7978 for Cell3. The complex Morlet 1.0 median increases from 0.6292 to 0.6772. The complex Morlet 1.5 median increases from 0.6128 to 0.6423, and the complex Morlet 3.0 median increases from 0.5373 to 0.5536. At periods of at least 2 Mpc, Cell3 remains higher for the complex Gaussian, Morlet 1.0, and Morlet 3.0 families. The galaxy-derived cell is higher by only about 0.0018 for Morlet 1.5. The substituted authoritative geometry therefore performs at least as well as the data-derived geometry and usually performs better.

9.3 Relation to the earlier 3-4 percent dilation result

The earlier 3-4 percent result measured a fine global dilation between a fixed galaxy-derived K-Field signal and the observed galaxy signal. The present 24.82 percent difference compares the equivalent-volume calibration scales of two different cell shapes. These are different quantities. A direct basis substitution changes edge ratios, plane geometry, node radii, and shell frequencies, so its independently fitted equivalent-volume scale need not remain within a few percent of the previous cell scale. The relevant result is that the calibrated Cell3 signal reaches stronger coherence under the same empirical test.

10. Ellipsoidal Shell Branch

The ellipsoidal experiment tests whether the shared organization is a generic consequence of any homothetic quadratic shell derived from Cell3 or whether the strongest relation is specifically radial about the observer. The ellipsoidal branch remains coherent, but the long-period response is substantially more dependent on wavelet family.

Wavelet	Full-band median	Median 1-20	Median 2-20	Maximum	Period at max
cmor1.0-1.0	0.596522	0.338785	0.295233	0.771974	0.600 Mpc
cmor1.5-1.0	0.529033	0.325143	0.280735	0.761045	0.600 Mpc
cmor3.0-1.0	0.415544	0.281156	0.233085	0.722127	0.600 Mpc
cgau2	0.787347	0.757737	0.579569	0.852624	3.274 Mpc

Table 11. Physical Cell3 ellipsoid wavelet-coherence summary.

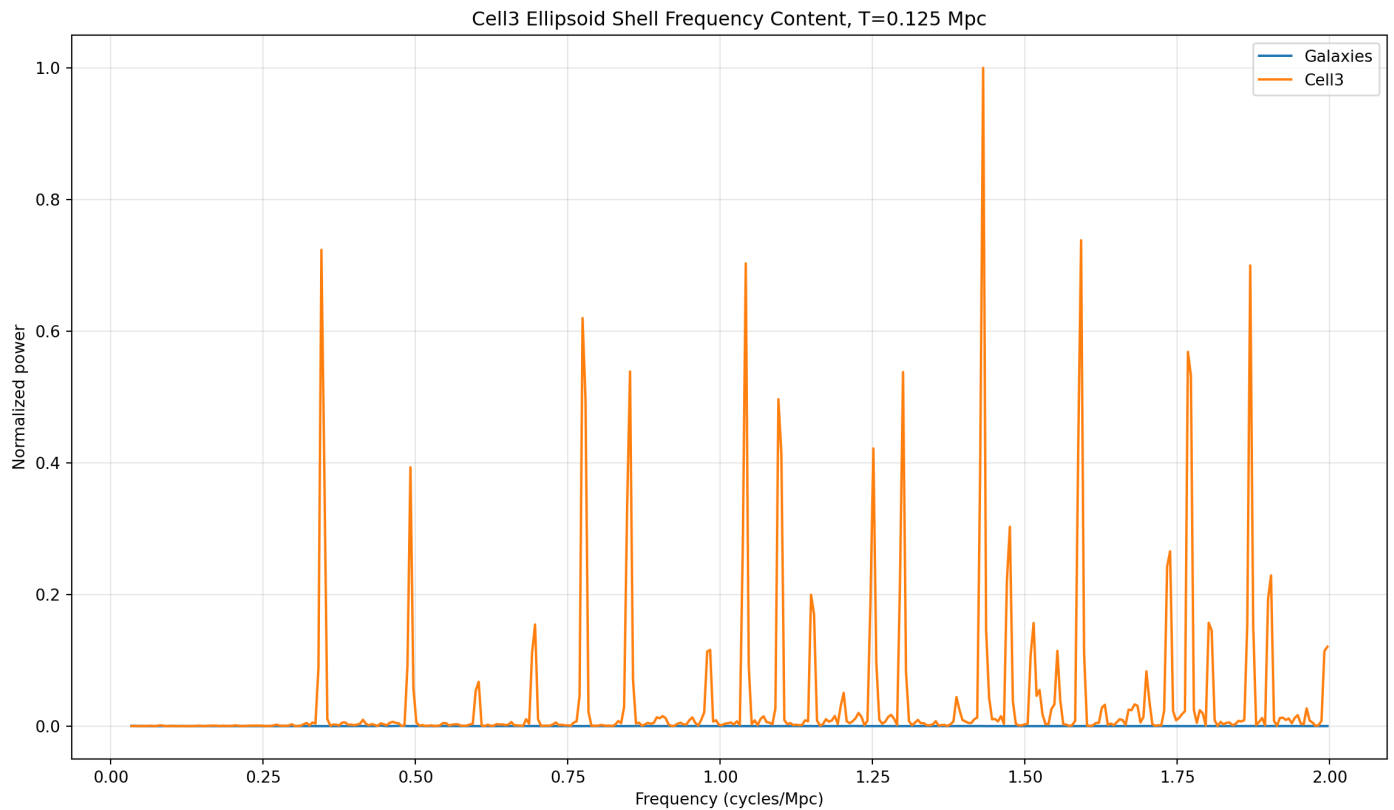


Figure 13. Fourier comparison for the physical Cell3 ellipsoid branch.

The complex Gaussian ellipsoid result remains strong, with a full-band median of 0.7873 and a 2-20 Mpc median of 0.5796. The Morlet long-period medians fall to 0.2331-0.2952. The spherical branch is therefore the cleaner multiwavelet witness. This geometry dependence constrains the interpretation: the observed common organization is principally radial about the observer rather than a generic response to every quadratic radius defined by Cell3.

11. Physical and Scientific Interpretation

11.1 What has been established

The Cell3 shape and ICRS orientation were fixed before astronomical calibration. The microscopic rendering scale was removed. One scalar was fitted by a full-length, multiwavelet shell-coherence procedure. The resulting dense, survey-masked node signal carries coherent radial structure with the actual 5,000-galaxy shell population. The same processing shows that Cell3 performs as well as or better than the galaxy-derived cell. This creates an independent bridge from the authoritative primitive K-Field geometry to a macroscopic astronomical population.

11.2 Why strong coherence can coexist with a noisy FFT

The galaxy series is sparse and distance-noisy. A narrow shell contains only a small number of objects, and catalog distance errors displace objects between neighboring shells. A global FFT requires stationary phase over the complete radial interval, so broadening and local displacement reduce its peak clarity. Wavelet coherence asks a less restrictive and more appropriate question: whether both signals contain corresponding period bands at corresponding radial locations. A sparse series can therefore retain high localized coherence even when its global Fourier spectrum is visually obscured.

11.3 Why weak cross-covariance does not negate coherence

Cross-covariance tests rigid translation. The galaxy and Cell3 shell signals are not rigid copies because the galaxy sample is incomplete, selection changes with distance, individual distances are uncertain, and coherent bands are intermittent. Wavelet coherence resolves common scale content without imposing one global lag. Strong coherence together with weak cross-covariance therefore identifies a multiscale relation, not a simple shifted waveform.

11.4 The calibrated Cell3 scale

The equivalent-volume scale of 46.1 Mpc is the empirical astronomical calibration of the determinant-one Cell3 shape. Its exact natural-gauge equivalent is a 126.318036607 Mpc body diagonal. The scale is not inferred from 1.3436536 nm and is not a conversion of nanometres into megaparsecs. It is obtained by maximizing a collective geometric correspondence between an independently fixed lattice shape and an observed shell population.

Principal finding. The correctly oriented Cell3 geometry, expressed in a unit-sphere natural gauge and calibrated by the galaxy shell witness, generates a spherical shell signal whose multiscale coherence is broad, persistent across wavelet families, and generally stronger than the result from the galaxy-derived cell.

12. New Discovery Pathways

12.1 Distance-uncertainty ensembles

Each galaxy distance can be represented by its reported uncertainty model or by a method-specific empirical error distribution. Repeated draws would produce an ensemble of shell signals and therefore a distribution of calibrated Cell3 scales and coherence maps. This experiment would determine how tightly the present 46.1 Mpc optimum is constrained by the catalog and whether the optimum is centered or skewed by nearby-galaxy distance errors.

12.2 Distance-method stratification

The catalog should be partitioned by distance method: primary standard candles, secondary indicators, group assignments, flow-corrected redshift distances, and uncorrected redshift proxies. A method with smaller radial error should produce sharper Cell3 coherence and a narrower scale optimum. The comparison becomes a geometric diagnostic of astronomical distance quality as well as a test of K-Field organization.

12.3 Larger and denser catalogs

The current sample contains 5,000 objects against millions of theoretical nodes. Increasing the galaxy population by one or two orders of magnitude will reduce shell shot noise, stabilize Fourier peaks, and permit thinner radial and angular sectors. The decisive requirement is documented distance provenance, not raw object count alone.

12.4 Sector and directional analysis

The complete sweep can be repeated by equal-area sky region. A uniform preferred scale would support a global radial calibration. Direction-dependent scale or phase would reveal anisotropic response tied to the ICRS orientation of Cell3, survey systematics, or local-flow distance corrections. Sector coherence also provides a direct bridge to the existing K-Field sky-response maps.

12.5 Depth persistence

The depth-4 node set should be compared with depths 3, 5, and higher. A genuine geometric shell family should exhibit predictable persistence, splitting, or harmonic refinement as dyadic depth increases. This test separates structural peaks from finite-depth aliasing and identifies which coherence bands belong to primitive Cell3 geometry versus recursive subdivision.

12.6 Cross-validation

The astronomical scale should be estimated on one galaxy subset and evaluated on an independent subset. The sky can be divided into disjoint sectors, or the catalog can be split by source and distance method. The scale is validated when it retains coherence without being re-fitted on the evaluation sample.

13. Complete Reconstruction Protocol

Step	Operation
1	Load the 5,000-galaxy table and convert each RA/Dec pair to an ICRS unit vector.
2	Construct the 72 by 36 RA/sin(Dec) occupancy mask from the actual galaxy directions.
3	Load the authoritative Cell3 ICRS basis and divide it by the body-diagonal norm 1.343653621680 to obtain the unit-sphere natural basis.
4	Compute $v_{\text{root}} = \det(B_{\text{natural}}) ^{1/3}$ and form $A_{\text{unit-volume}} = B_{\text{natural}}/v_{\text{root}}$.
5	Set the Cell3 phase and origin to zero. Do not import the galaxy-derived phase.
6	For each trial scale S , generate depth-4 positions $x = S A_{\text{unit-volume}}(m/16)$ over an integer range covering the radial domain.
7	Discard nodes outside 13.327534876-218.674365624 Mpc and apply the galaxy survey mask to the remaining directions.
8	Form spherical shell counts on a 0.05 Mpc radial grid at $T=0.125, 0.25, 0.5, 1.0$, and 2.0 Mpc.
9	Divide each shell count by $4\pi r^2$ and normalize the signal for the selected analysis.
10	Evaluate four-wavelet full-length coherence against the galaxy signal and select the consensus scale.
11	At the calibrated scale, compute complete coherence maps, radially averaged coherence, detrended Hann-window FFT spectra, and normalized cross-covariance.
12	Construct $H = (A A^T)^{-1}$ and repeat the matched analysis for the physical ellipsoid branch.
13	Run the identical-method comparison with the previously calibrated galaxy-derived cell.
14	Record all arrays, figures, scale tables, and reconstruction constants in machine-readable form.

Table 12. End-to-end reconstruction procedure.

14. Conclusions

The K-Field Cell3 and galaxy shell experiment creates a direct, reproducible comparison between an authoritative geometric lattice and an independent astronomical population. Cell3 is used in its correct ICRS orientation. Its microscopic rendering magnitude is removed, its body diagonal is normalized to the unit-sphere natural gauge, and its determinant is converted to a unit-volume calibration gauge. A single astronomical scale is then obtained from the galaxy shell witness. The resulting scale is 46.1 Mpc in equivalent-volume units, or 126.318036607 Mpc along the natural Cell3 body diagonal. The calibrated depth-4 construction produces nearly thirteen million nodes over the galaxy radial domain and more than 6.88 million nodes after matched survey masking. The spherical shell signal exhibits strong full-length wavelet coherence with the 5,000-galaxy signal. The complete-band median ranges from 0.5536 to 0.7978 across four distinct wavelets. Fourier analysis identifies exact and near-matching radial periods, while cross-covariance remains weak, demonstrating that the relationship is multiscale and localized rather than a rigid translation. The physical ellipsoid branch retains coherence but is more wavelet-family dependent, establishing the spherical radial construction as the clearer witness. Under an identical analysis method, Cell3 exceeds the galaxy-derived cell in every complete-band median and in most long-period medians. This result changes the role of galaxy data in K-Field analysis. The galaxy population is no longer used only to reconstruct an auxiliary cell. It independently calibrates and validates the shell organization generated by Cell3 itself. The agreement survives severe population sparsity, nonuniform survey support, and approximate catalog distances. The completed work therefore establishes a new pathway for geometric cosmological analysis: dense deterministic field structure can be tested against sparse astronomical populations through matched masks, shell statistics, scale calibration, and localized coherence.

Final statement. The unit-sphere natural Cell3 geometry, fixed in ICRS and independently calibrated at astronomical scale, is a stronger shell-coherence witness for the observed 5,000-galaxy radial population than the previously reconstructed galaxy-derived cell under the same computational standard.

Appendix A. Complete Cell3 Numerical Kernel

Item	Value
B_render body diagonal	1.343653621679882
Natural determinant	0.048607879555281
Determinant cube root	0.364951840910143
Unit-volume determinant	0.999999999999999
Phase	(0, 0, 0)
Calibrated equivalent-volume scale	46.100000000000 Mpc
Natural body-diagonal scale	126.318036607330 Mpc
Sheet normal	(-0.876976866366, -0.047223124230, 0.478206600118)
Sheet normal RA/Dec	183.082265054 deg, 28.568337611 deg

Table A1. Minimal numerical constants required to reconstruct the Cell3 astronomical geometry.

Edge	x (Mpc)	y (Mpc)	z (Mpc)	Length	RA	Dec
a	30.788674415	-5.160915090	-3.524232501	31.416520042	350.484328	-6.440865
b	3.321476729	19.490318066	-51.461648469	55.128993907	80.328749	-68.983472
c	-0.952265892	-36.644278303	-64.842527002	74.486664919	268.511404	-60.519741

Table A2. Calibrated Cell3 ICRS basis vectors.

Appendix B. All 27 Centered Half-Step Lucas Nodes

Each node is defined by integer triplet (i,j,k) in $\{-1,0,1\}^3$ and position $r=B(i,j,k)/2$. The sheet index shown below is $\lambda=i+j+k$. Natural coordinates use the unit-body-diagonal gauge; calibrated coordinates use the independently fitted Mpc geometry.

Node	i	j	k	lambda	x natural	y natural	z natural
N-1-1-1	-1	-1	-1	-3	-0.131247628	0.088328143	0.474312344
N-1-1+0	-1	-1	0	-2	-0.135016946	-0.056719544	0.217648574
N-1+0-1	-1	0	-1	-2	-0.118100349	0.165475947	0.270613609
N+0-1-1	0	-1	-1	-2	-0.009377959	0.067899885	0.460362505
N-1-1+1	-1	-1	1	-1	-0.138786265	-0.201767232	-0.039015196
N-1+0+0	-1	0	0	-1	-0.121869668	0.020428259	0.013949839
N-1+1-1	-1	1	-1	-1	-0.104953071	0.242623750	0.066914874
N+0-1+0	0	-1	0	-1	-0.013147278	-0.077147803	0.203698735
N+0+0-1	0	0	-1	-1	0.003769319	0.145047688	0.256663770
N+1-1-1	1	-1	-1	-1	0.112491709	0.047471626	0.446412666
N-1+0+1	-1	0	1	0	-0.125638987	-0.124619429	-0.242713931
N-1+1+0	-1	1	0	0	-0.108722390	0.097576062	-0.189748896
N+0-1+1	0	-1	1	0	-0.016916597	-0.222195491	-0.052965035
N+0+0+0	0	0	0	0	0.000000000	0.000000000	0.000000000
N+0+1-1	0	1	-1	0	0.016916597	0.222195491	0.052965035
N+1-1+0	1	-1	0	0	0.108722390	-0.097576062	0.189748896
N+1+0-1	1	0	-1	0	0.125638987	0.124619429	0.242713931
N-1+1+1	-1	1	1	1	-0.112491709	-0.047471626	-0.446412666
N+0+0+1	0	0	1	1	-0.003769319	-0.145047688	-0.256663770
N+0+1+0	0	1	0	1	0.013147278	0.077147803	-0.203698735
N+1-1+1	1	-1	1	1	0.104953071	-0.242623750	-0.066914874
N+1+0+0	1	0	0	1	0.121869668	-0.020428259	-0.013949839
N+1+1-1	1	1	-1	1	0.138786265	0.201767232	0.039015196
N+0+1+1	0	1	1	2	0.009377959	-0.067899885	-0.460362505
N+1+0+1	1	0	1	2	0.118100349	-0.165475947	-0.270613609
N+1+1+0	1	1	0	2	0.135016946	0.056719544	-0.217648574
N+1+1+1	1	1	1	3	0.131247628	-0.088328143	-0.474312344

Table B1. Complete 27-node natural-coordinate list.

Node	x (Mpc)	y (Mpc)	z (Mpc)	Radius (Mpc)	Sheet
N-1-1-1	-16.578943	11.157438	59.914204	63.159018	1
N-1-1+0	-17.055076	-7.164701	27.492940	33.137144	2
N-1+0-1	-14.918204	20.902597	34.183380	42.754822	2
N+0-1-1	-1.184605	8.576980	58.152088	58.793139	2
N-1-1+1	-17.531209	-25.486841	-4.928323	31.324283	3
N-1+0+0	-15.394337	2.580458	1.762116	15.708260	3
N-1+1-1	-13.257466	30.647756	8.452556	34.445479	3
N+0-1+0	-1.660738	-9.745159	25.730824	27.564497	3
N+0+0-1	0.476133	18.322139	32.421264	37.243332	3
N+1-1-1	14.209732	5.996523	56.389971	58.461129	3
N-1+0+1	-15.870470	-15.741682	-30.659147	37.942795	4
N-1+1+0	-13.733599	12.325617	-23.968708	30.249488	4
N+0-1+1	-2.136871	-28.067298	-6.690439	28.932705	4
N+0+0+0	0.000000	0.000000	0.000000	0.000000	4
N+0+1-1	2.136871	28.067298	6.690439	28.932705	4
N+1-1+0	13.733599	-12.325617	23.968708	30.249488	4
N+1+0-1	15.870470	15.741682	30.659147	37.942795	4
N-1+1+1	-14.209732	-5.996523	-56.389971	58.461129	5
N+0+0+1	-0.476133	-18.322139	-32.421264	37.243332	5
N+0+1+0	1.660738	9.745159	-25.730824	27.564497	5
N+1-1+1	13.257466	-30.647756	-8.452556	34.445479	5
N+1+0+0	15.394337	-2.580458	-1.762116	15.708260	5
N+1+1-1	17.531209	25.486841	4.928323	31.324283	5
N+0+1+1	1.184605	-8.576980	-58.152088	58.793139	6
N+1+0+1	14.918204	-20.902597	-34.183380	42.754822	6
N+1+1+0	17.055076	7.164701	-27.492940	33.137144	6
N+1+1+1	16.578943	-11.157438	-59.914204	63.159018	7

Table B2. Complete 27-node calibrated physical-coordinate list.

Appendix C. Machine-Readable Evidence Files

File	Role
cell3_shell_coherence_summary.json	Complete geometry, calibration, spherical, ellipsoidal, and comparison summary.
cell3_scale_sweep_coarse.csv	Coarse equivalent-volume scale search.
cell3_scale_sweep_consensus.csv	Refined four-wavelet consensus around candidate peaks.
cell3_galaxy_spherical_sweep.csv	All radial spherical shell curves at five shell half-widths.
cell3_galaxy_ellipsoid_sweep.csv	All physical ellipsoid shell curves.
cell3_galaxy_wavelet_summary.csv	Spherical wavelet summary statistics.
cell3_galaxy_ellipsoid_wavelet_summary.csv	Ellipsoid wavelet summary statistics.
cell3_galaxy_global_coherence_by_period.csv	Period-dependent spherical coherence.
cell3_galaxy_fft_spectra.csv	Complete Fourier spectra.
cell3_galaxy_fft_peaks.csv	Leading spectral peaks and comparisons.
cell3_galaxy_xcov.csv	Cross-covariance curves by lag.
cell3_galaxy_xcov_summary.csv	Cross-covariance extrema and zero-lag values.
cell3_vs_galaxy_cell_wavelet_comparison.csv	Complete identical-method geometry comparison.
cell3_vs_galaxy_cell_wavelet_summary.csv	Condensed comparison table.

Table C1. Machine-readable source and evidence files used by this report.

Archival Notice

This document contains proprietary confidential information and trade secrets of Krytronx, LLC. It is provided as an archival scientific and educational reference under the direction of P. Dwayne Esterline, Principal Investigator. All rights are reserved. Distribution, reproduction, disclosure, or derivative use is prohibited except by explicit authorization of Krytronx, LLC.

Document filename: K-Field_cel3_and_Galaxy_shell_coherence_1784030762.pdf

Publication epoch: 1784030762

Geometry: authoritative Cell3 in ICRS unit-sphere natural and unit-volume calibration gauges

Empirical witness: 5,000-galaxy RA/Dec/distance catalog

Primary calibrated scale: 46.100000 Mpc equivalent-volume scale

Primary natural-gauge diagonal: 126.318036607 Mpc